# TITAN IPsec Security Intelligence Platform

CYBERSECURITY & COMPLIANCE AUDIT REPORT

**Report ID: REP-20260829175305-1e4ad3**
**Target File: ipsec_test.pcap**

**Compliance: COMPLIANT (Adequate Protection)**
**Timestamp: Aug 29, 2026, 17:53 UTC**

## 1. EXECUTIVE SECURITY SUMMARY & POSTURE GRADE

**GRADE A** **Normalized Risk Index: 30 / 100 (LOW RISK)** *Status: Verified by TITAN Engine*

## 2. CRYPTOGRAPHIC INTEGRITY & POST-QUANTUM (PQC) READINESS

- Payload Encryption: Enforced (ESP Protocol 50)
- Authentication Header: Present (AH Protocol 51)
- Active SPIs: 0x1001, 0x1002
- Post-Quantum Readiness: 30% (QUANTUM-VULNERABLE)
- IKE PSK Exposure Risk: LOW (No IKEv1 Aggressive Mode Hash Exposure)
- CNSA 2.0 Compliance: UPGRADE REQUIRED

## 3. TRAFFIC TELEMETRY & PROTOCOL DISTRIBUTION MATRIX

| Protocol Layer / Name | Packet Count | Traffic Share (%) | Security Classification |
|---|---|---|---|
| ESP Payload Encryption (Protocol 50) | 1 | 33.3% | Encrypted (Secure) |
| Authentication Header AH (Protocol 51) | 1 | 33.3% | Cleartext / Control |
| IKE / NAT-T (UDP Port 500 / 4500) | 1 | 33.3% | Cleartext / Control |
| TCP Transport Streams | 0 | 0.0% | Cleartext / Control |
| UDP Datagrams (Non-VPN) | 0 | 0.0% | Cleartext / Control |
| DNS Resolution Queries | 0 | 0.0% | Cleartext / Control |
| ICMP Diagnostic Messages | 0 | 0.0% | Cleartext / Control |
| **TOTAL CAPTURED TRAFFIC** | **3** | **100.0%** | **Full Network Stream** |

## 4. ENCRYPTED TRAFFIC ANALYSIS (ETA) & APPLICATION FINGERPRINTING

- Inferred Application: Interactive Shell / VoIP / Command Stream
- Traffic Pattern: Low Latency / Fixed Size Frames
- ETA Confidence Score: 94.5%
- Burstiness Index: 0.03

Behavioral Heuristic: Continuous stream of small, fixed-size encrypted frames indicative of interactive management sessions or voice-over-IP.

## 5. MITRE ATT&CK ENTERPRISE FRAMEWORK MAPPING

| Technique ID | Technique Name | Tactic / Domain | Severity | Audit Finding |
|---|---|---|---|---|
| T1040 | Network Sniffing | Credential Access / | **HIGH** | AH Protocol 51 exposes plaintext |

## 6. DEEP SECURITY AUDIT FINDINGS

[1] ESP Payload Encryption Verified: 1 encrypted frames (33.3% of capture).
[2] IKE Handshake Observed: 1 key exchange negotiation packets detected on UDP 500/4500.
[3] Authentication Header (AH) Active: 1 packets use Protocol 51. Note: AH does NOT provide data confidentiality/encryption.
[4] Zero Cleartext Leakage: No plaintext transport protocols were observed bypassing the VPN tunnel.
[5] MTU Headroom Optimal: Average packet length is 42 bytes (Max: 44 bytes), avoiding fragmentation.
[6] Anti-Replay Protection: Sequence tracking verified across 2 active Security Associations (SAs).

## 7. ACTIONABLE HARDENING ROADMAP & SIEM CONFIGURATION

**Step 1:** Migrate from AH (Protocol 51) to ESP (Protocol 50) with AEAD ciphers (e.g. AES-GCM-256) for data confidentiality.